Modification of Fisher's Geometric Model - flood depiction

Dominika Rawa, Maja Domańska

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Chapter 1

Introduction

Over time, populations adapt to surrounding them environments, steadily evolving. In this study the differences between a pre-adapted in a steadily changing environment and un-adapted populations' behaviour are analysed. Particularly, a case of a flood appearing repetitively every couple of years is observed. In that scenario a bottleneck effect, which sharply changes the optimum, occurs. The aftermath of the catastrophy is the main subject of the study. The research question is: Is a population that has adapted to gradual linear drift more or less resilient to a sudden large displacement of the optimum than a naive (un-adapted) population?

Between a pre-adapted and an un-adapted populations, the unadapted one may be more resilient to a catastrophic event, a drift. High variance in freshly initialised groups might make them less fragile in case of a sudden environmental change. A lower mortality rate is expected to appear in the un-adapted populations.

Chapter 2

Methods

Two independent populations were initialised: a pre-adapted and a un-adapted (naive) one. Naive population was initialised in the flood environment. The pre-adapted population had been adapting to the baseline environment (global warming) for 60 generations(*pre_adaptation_generations* parameter) before being put in the flood environment for 400 generations (*flood_generations* parameter) along with the naive population. In every population there were 100 individuals (N parameter).

Twenty independent replicates and delta$\in \{0, 0.01\}$ were used in the simulation. Delta represents stochastic noise added to drift

Fischer's geometric model assuming asexual cloning, isotropic mutation, and a linearly drifting optimum was modified in order to accomodate for floods. In this model, the environment defines an optimal phenotype $\alpha(t)$ at each generation. Fitness (φ) is a Gaussian function of phenotypic distance from that optimum. A perfect match gives $\varphi = 1$; fitness decays exponentially with distance. The width σ controls how strict the environment is — smaller σ means only near-perfect phenotypes survive. The optimal phenotype is alpha, alpha0 is the starting optimal phenotype.

2.1 Phenotype

Every individual is described with a 4-dimensional vector of features, all of which are drawn from the Gaussian distribution. Those are: ability to find food, ability to swim and the organism's (x,y) coordinates. While in the baseline scenario all feature optima drift steadily in a fixed direction with small stochastic fluctuations, in a flood-endangered environment only ability to find food behaves that way.

Ability to swim is a scalar between [-0.5, 0.5]. After being drawn from the Gaussian distribution, it is then clipped to match the set scale in which 0.5 means excellent swimming skills and is therefore the optimum in case of a flood. In baseline environment, the optimum is set to 0.0.

It is assumed that every individual's location can be described by (x,y) Cartesian coordinates. The optimal position is fixed at hill top, (0,0) point at all times and is unaffected by environmental shocks.

2.2 Environment

2.2.1 baseline environment

Baseline environment is an environment in which the optimum drifts steadily in a fixed direction with small stochastic fluctuations:

$$\alpha(t+1) = \alpha(t) + N(c, \sigma^2 I)$$

where c is the mean drift per generation ([0.01, 0.01, 0.01, 0.01] and σ adds noise. That simulates the global warming. No sudden events happen, directional adaptation occurs.

2.2.2 flood environment

While the linear shift and noise in the environment are still present, they do not change the location or the ability to swim.

Floods that occur every 40 (*T_shock* parameter) years are introduced. The optimum for swimming skills is 0.0 normally. With a flood, it changes to 0.5. The optimum for location is fixed at [0.0, 0.0]. In case of a flood, the optimum for ability to find food jumps by a random displacement drawn from N(0, sigma_shock^2 I). sigma_shock = 0.03 is used.

The *sigma_shock* parameter controls the standard deviation of the optimum's displacement.

2.3 statistical tests

Statistical analysis of survival rates was performed using Fisher's exact test to assess differences between conditions.

In the no-noise scenario (delta = 0), the baseline condition showed complete survival (20/20), compared to naive flood (15/20) and pre-adapted flood (7/20). All differences between conditions are statistically significant, with small p-value. For baseline and naive flood $p = 0.047$, for naive flood and pre-adapted $p = 0.0248$, and for baseline and pre-adapted flood $p < 0.001$. confirming that pre-adaptation was strongly disadvantageous under stable conditions.

In the presence of environmental noise (delta = 0.02), survival differed across conditions (naive flood: 16/20, pre-adapted flood: 10/20, baseline: 6/20). The difference between baseline and naive flood was statistically significant (p = 0.0036). However, differences between naive flood and pre-adapted flood (p = 0.0958) and between baseline and pre-adapted flood (p = 0.3332) were not statistically significant. Thus, although differences in survival are observed, the

p-values indicate that some of them may be due to random variation rather than a true underlying effect.

Chapter 3

Results

3.1 Plots

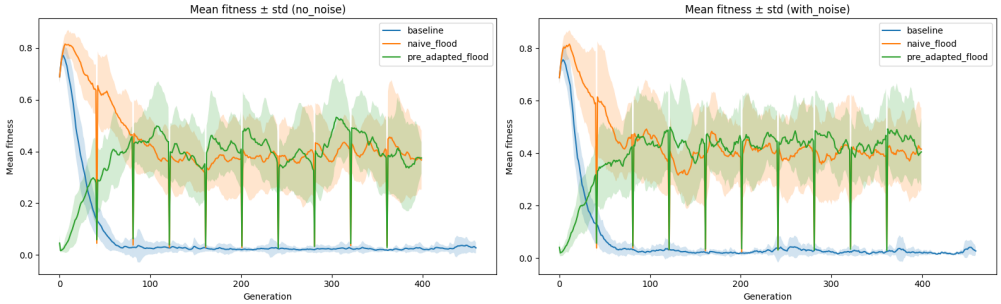

Figure 3.1: Mean fitness trajectory
Initially mean fitness of the naive population is very high (0.8) and the pre-adapted one is low (about 0). After two floods, the values become similar. The population that has slightly better fitness changes every second flood. With every catastrophe mean fitness drastically declines. At first, higher variance in naive population results in mean of fitness being significantly higher than in the pre-adapted population. That advantage does not last for the whole experiment.

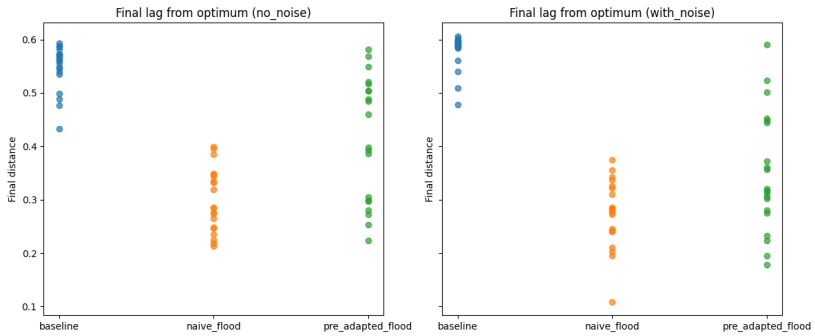

Figure 3.2: Lag from the optimum
Statistical dispersion is higher in the pre-adapted population than in the naive one. While the naive population has the lowest lag from optimum, the baseline has the highest. The naive population is significantly better adapted to flood environment. In general, it is better adapted to flooding than the pre-adapted population.

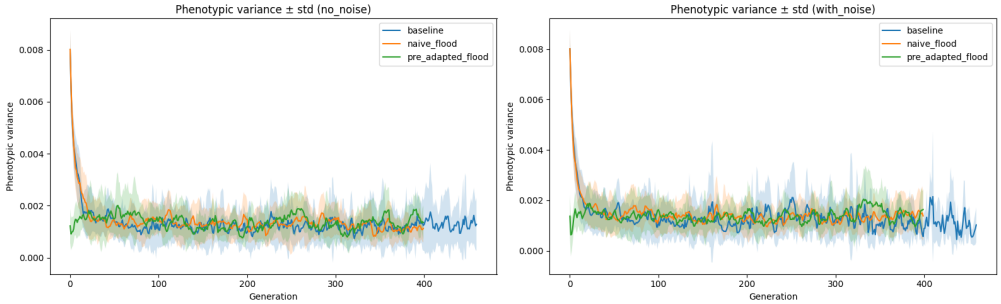

Figure 3.3: Phenotypic variance
The naive population has a high variance that sharply drops at the very start of the experiment. The rest of the populations behave similarily.

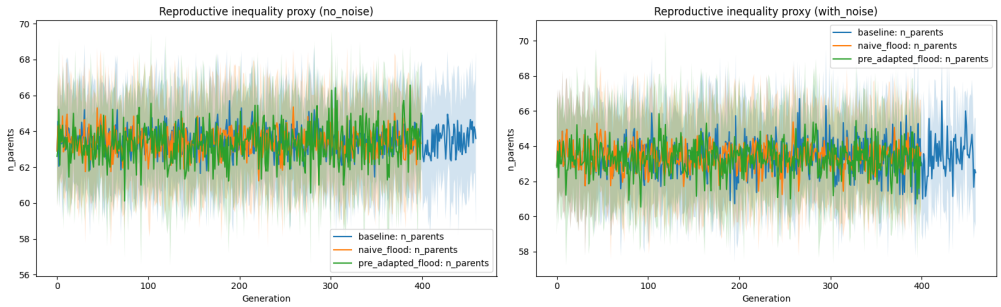

Figure 3.4: Reproductive inequality
Reproductive inequalities througout researched populations show no noticeable differences between them that could be informative in this study.

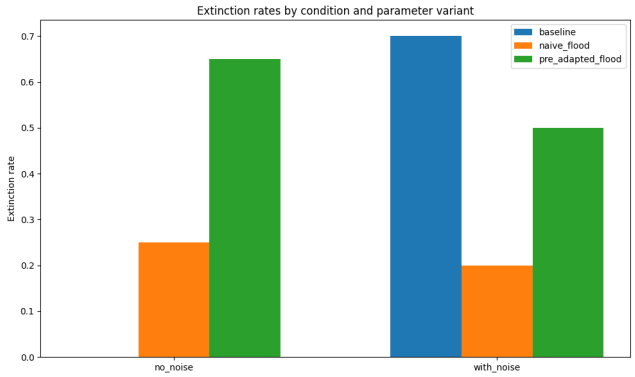

Figure 3.5: Extinction rate
Baseline scenario extinction rates are significantly higher with noise added. The rest of the scenarios' rates drop slightly with added noise. both naive and pre-adapted populations, adding a noise lowers extinction rates.

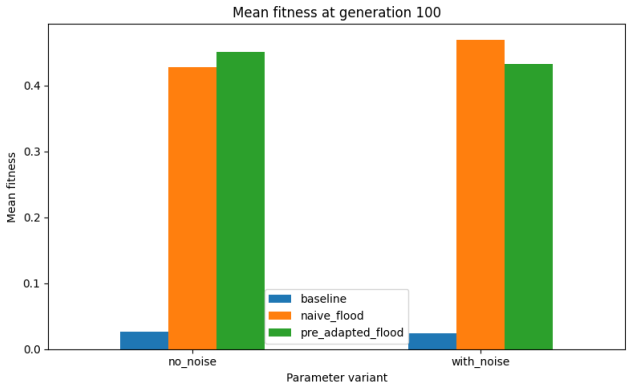

Figure 3.6: Parameter sensitivity
Introducing environmental noise caused the naive and the pre-adapted populations to reverse their mean fitness scores, but no effect was visible on the baseline population.

Of all figures above, the most informative are: Mean fitness trajectory, Lag from the optimum and parameter sensitivity. The first one shows the change in populations in time. The lag plot provides a visual and easy to read representation of distance from the optimum that allows the reader to interpret the differences between researched populations. Parameter sensitivity allows to see the differences that added environmental noise makes.

3.2 Interpretation

Initially, the naive population shows high mean fitness due to it's large variance in phenotypes that increases the probability of a randomly chosen individual being well adapted to rapidly changing environment.

After two flood events, the differences between the two populations weaken. With every flood, the dominating strategy changes.

The lag from optimum clearly indicates that the naive population is better adapted to the unstable environment.

The variance trajectories show a sharp drop in variance with each flood in both populations. Lowered variances then stay on similar level for all groups.

Extinction rates show that added noise allows populations to improve their performance in flooding environment by not allowing the group lower their phenotype variance too much and, in consequence, become too narrowly specialized.

Chapter 4

Discussion

Overall, the naive population is better adapted to the flood environment, compared to the pre-adapted population. The results match the prediction.

Over time, population evolves. Sustained directional selection reduces phenotypic variance — the population "specialises" in one direction. The adaptations allow the population to become well adapted to stable environment. When the conditions change drastically, the population with higher variance is advantaged. That is because in that population there is a broader range of traits. In turn, the probability of an individual "randomly" being better adapted to such environment is higher.

In this setting, further research of the way sexual reproduction would affect those populations might prove insightful.